

Q1. Which of the following is a function?

- (A) $y = x^2$
 (B) $x = y^2$
 (C) $x^2 + y^2 = 1$
 (D) $y = \pm x$
 (E) $y = \sqrt{x}$

Q2. What is the domain of the function $f(x) = \frac{1}{x}$?

- (A) $(-\infty, \infty)$
 (B) $(-\infty, 0) \cup (0, \infty)$
 (C) $[0, \infty)$
 (D) $(-\infty, 0]$
 (E) $[0, \infty)$

Q3. Which of the following relations is not a function?

- (A) $y = x$
 (B) $y = x^2$
 (C) $x = y^2$
 (D) $y = \pm x$
 (E) $y = x^3$

Q4. What is the range of the function $f(x) = x^2$?

- (A) $(-\infty, \infty)$
 (B) $[0, \infty)$
 (C) $(-\infty, 0]$
 (D) $[0, \infty)$
 (E) $(-\infty, 0)$

Q5. Evaluate $f(-2)$ if $f(x) = x^2 - 3x + 2$

- (A) 0
 (B) -2
 (C) -4
 (D) 4
 (E) 6

Q6. What is the domain of the function $f(x) = \sqrt{x}$?

- (A) $(-\infty, \infty)$
 (B) $[0, \infty)$
 (C) $(-\infty, 0)$
 (D) $(-\infty, 0]$
 (E) $[0, \infty)$

Q7. Which of the following functions represents a vertical line?

- (A) $y = x$
 (B) $y = x^2$
 (C) $x = 2$
 (D) $y = \sqrt{x}$
 (E) $y = \frac{1}{x}$

Q8. What is the range of the function $f(x) = \sqrt{x}$?

- (A) $(-\infty, \infty)$
 (B) $[0, \infty)$
 (C) $(-\infty, 0]$
 (D) $(-\infty, 0)$
 (E) $[0, \infty)$

Open Ended Questions (FRQ)**Q9.** Determine whether the relation $y = \pm\sqrt{x}$ is a function. Explain your reasoning and provide a graph to support your answer.**Q10.** Find the domain and range of the function $f(x) = \frac{1}{x^2-4}$. Provide a graph to support your answer.**Chef's Tips**

- Check for any misconceptions about functions and relations
- Emphasize the importance of the vertical line test
- Make sure to provide clear and concise feedback